

# An Introduction to Climate Modeling

## Milestone 1 - Geography and Visualization

### 1 Read-in and Plot Geography

In this milestone, the goal is to read-in the geography information of planet Earth and plot it as a two-dimensional world map. We consider an equirectangular grid of the Earth, i.e., an equidistant rectangular grid in spherical coordinates, where the grid point  $(i, j)$  has the spherical coordinates  $(\theta_i, \varphi_j)$ , where  $\theta_i$  is the latitude between  $-90^\circ$  south and  $90^\circ$  north (including the poles) and  $\varphi_j$  the longitude between  $-180^\circ$  west and  $180^\circ$  east. The basis for this is the input file `The_World65x128.dat`, which describes the distribution of the different Earth surface types. This file contains a matrix  $G \in \mathbb{N}^{65 \times 128}$  with entries  $g_{ij} \in \{1, 2, 3, 5\}$ , where the entry  $g_{ij}$  stores the Earth surface type at grid point  $(i, j)$ . Here, 1 represents the Earth surface type *land*, 2 represents *sea ice*, 3 represents *snow*, and 5 represents *ocean*. The grid resolution in longitude and latitude direction is  $2.8125^\circ$ . The basis for this distribution and grid is from Zhuang et al.<sup>1</sup> You can proceed as follows:

1. Write a function `read_geography`, which reads the file `The_World65x128.dat` from the folder `input` and outputs a matrix  $T \in \mathbb{N}^{65 \times 128}$  with the classification of the earth surface types.
2. Write a function `robinson_projection`, which maps an equirectangular grid in spherical coordinates to the plane. For simplicity use the approximate formula by Beineke for the Robinson projection,

$$x(\theta, \varphi) = \frac{\varphi}{\pi} (0.0379\theta^6 - 0.15\theta^4 - 0.367\theta^2 + 2.666),$$

$$y(\theta, \varphi) = 0.96047\theta - 0.00857 \operatorname{sign}(\theta) |\theta|^{6.41},$$

where  $\varphi$  is the longitude and  $\theta$  the latitude in radians. This function should return two matrices  $X = x_{ij}$  and  $Y = y_{ij}$ , where  $x_{ij} = x(\theta_i, \varphi_j)$ ,  $y_{ij} = y(\theta_i, \varphi_j)$ .

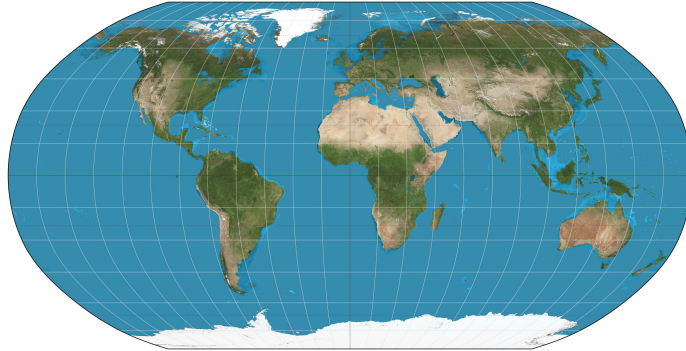


Figure 1: Robinson projection of the world <sup>2</sup>

3. Write a function `plot_geo` that creates a plot of the Earth surface type  $g_{ij}$  against the mapped coordinates  $(x_{ij}, y_{ij})$ .
4. Use these functions in a program and run it to check your results.

<sup>1</sup>K. Zhuang, G.R. North, M.J. Stevens, *A NetCDF version of the two-dimensional energy balance model based on the full multigrid algorithm*, SoftwareX, Vol. 6, pp. 198-202, July 7, 2017.

<sup>2</sup>Daniel R. Strebe, [https://en.wikipedia.org/wiki/File:Robinson\\_projection\\_SW.jpg](https://en.wikipedia.org/wiki/File:Robinson_projection_SW.jpg)

## 2 Bonus

1. Implement the Winkel tripel projection, as an alternative to the Robinson projection, given by

$$\begin{aligned}x(\theta, \varphi) &= \frac{1}{2} \left( \varphi \cos(\theta_1) + \frac{2 \cos(\theta) \sin(0.5\varphi)}{\text{sinc}(\alpha)} \right), \\y(\theta, \varphi) &= \frac{1}{2} \left( \theta + \frac{\sin(\theta)}{\text{sinc}(\alpha)} \right), \\ \alpha(\theta, \varphi) &= \arccos(\cos(\theta) \cos(0.5\varphi)),\end{aligned}$$

where  $\theta_1$  is the standard parallel and sinc is the unnormalized sine function that are

$$\theta_1 = \arccos\left(\frac{2}{\pi}\right) \quad \text{and} \quad \text{sinc}(x) = \begin{cases} \frac{\sin(x)}{x} & x \neq 0 \\ 1 & x = 0 \end{cases},$$

respectively.

2. Write a function to read in other data sets, like ICE-5G<sup>3</sup>, originally presented and analyzed by Peltier<sup>4</sup>. This function should parse this (binary) data which is provided in netCDF format. The data is given on an enriched grid, e.g., with  $1^\circ$  resolution ( $360 \times 180$  points) or  $10'$  resolution ( $2160 \times 1080$  points). So, this read-in routine should also have the option to downsample the data in some way while maintaining an equirectangular grid in lat/lon coordinates.

After readin (and downsampling), write another routine that can categorize the Earth surface types  $g_{ij} \in \{1; 2; 3; 5\}$  based on RGB color values for plotting. The *land* = 1, *ice* = 3, and *ocean* = 5 masking can be done in a straightforward way from the *orog* data is the surface elevation which is positive for land and negative for ocean. Also, the *sftgit* and *sftgif* data provides information about the land ice sheets. Sea ice is trickier to determine and one needs to combine some strategy with CLIMAP to determine the sea ice. This is how the original data from Zhuang et al. created their data.

Note, that the netCDF format may store the latitude and longitude in a different orientation than expected, e.g., South to North rather than North to South. Thus, some reordering of the points is necessary.

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<sup>3</sup><https://www.atmosp.physics.utoronto.ca/peltier/data.php>

<sup>4</sup>Peltier, W.R., Global glacial isostasy and the surface of the ice-age Earth: the ICE-5G (VM2) model and GRACE. Annual Review of Earth and Planetary Sciences, May 2004, Vol 32, 111-149.

### 3 Control Solutions

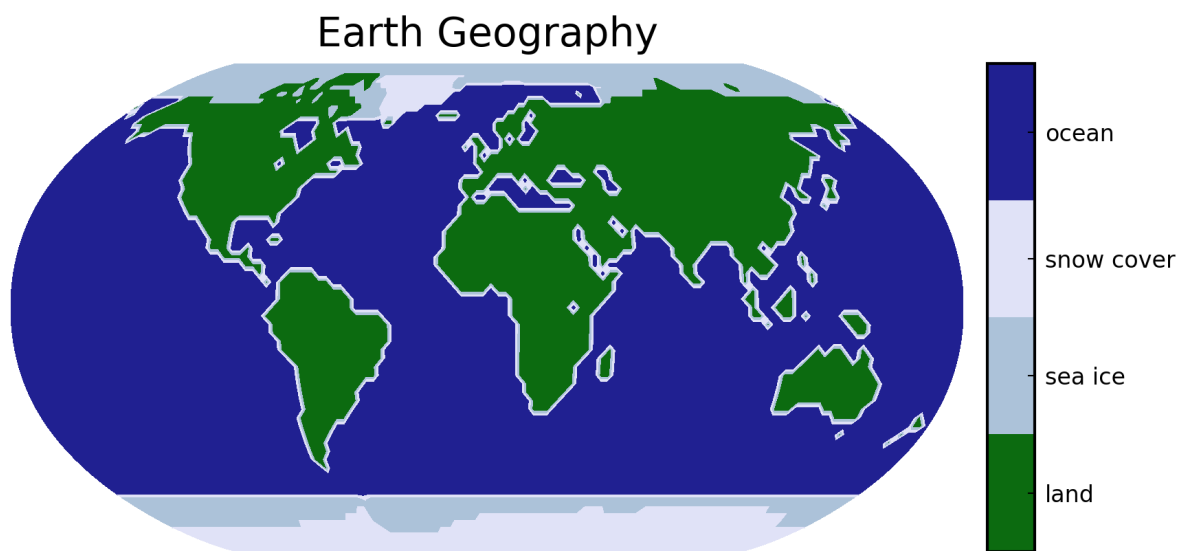


Figure 2: Robinson projection output